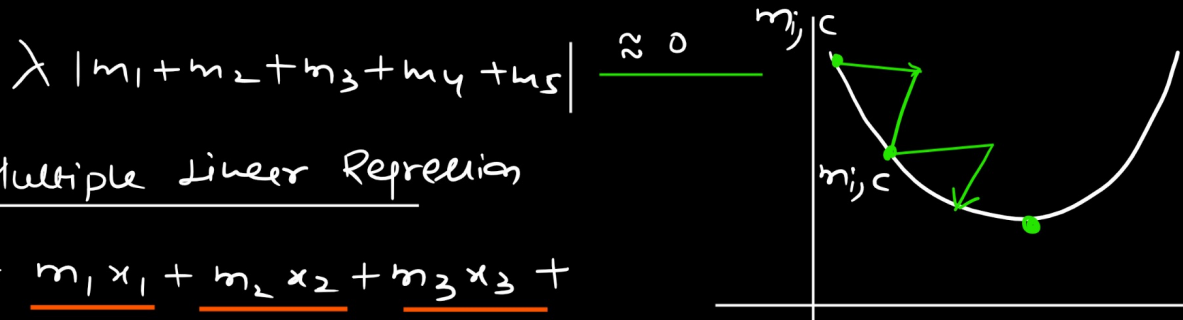


Regularization Technique

① Lasso Regression (L1 Regularization) slope/coefficients

hyperparameter cost function = training set
$$\text{MSE} \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right] + \lambda \sum_{i=1}^n |m_i|$$



Multiple Linear Regression

$$\hat{y} = \underline{m_1 x_1} + \underline{m_2 x_2} + \underline{m_3 x_3} + \underline{m_4 x_4} + \underline{m_5 x_5} + c$$

$\{m_1, m_2, m_3, m_4, m_5\} \rightarrow$ slope/coefficients

$\{x_1, x_2, x_3, x_4, x_5\} \rightarrow$ input/independent

feature

$x_2, x_4 \begin{cases} m_2 \approx 0 \\ m_4 \approx 0 \end{cases} \leftarrow \begin{matrix} \text{After some} \\ \text{iteration} \end{matrix}$

they are not important

$$m_{\text{new}} = m_{\text{old}} - \eta * \frac{\delta L}{\delta m}$$

feature

$$c_{\text{new}} = c_{\text{old}} - \eta * \frac{\delta L}{\delta c}$$

to consider

$\rightarrow$ feature selection

technique

↳ overfitting
reduce by
enhancing the
training time
↳ feature
selection

$$y = \underline{2x_1 + 3x_2 + 7x_3 + 9x_4 + c}$$

$\underline{2^2 + 3^2 + 7^2 + 9^2}$

② Ridge Regression (L2 Regularization)

$\lambda = 1$

Cost
function



$$MSE + \lambda \sum_{i=1}^n m_i^2$$

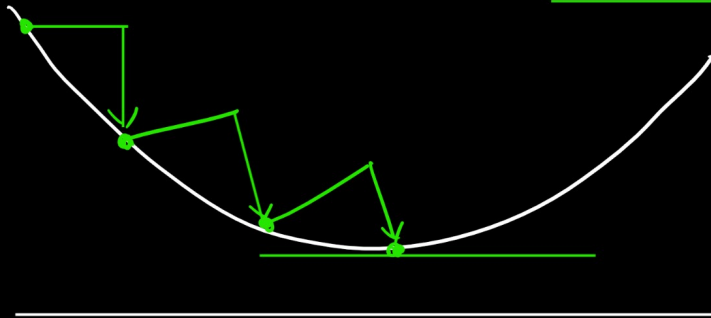
hyperparameter

$$m_1^2 + m_2^2 + m_3^2 + m_4^2 +$$

$$m_5^2$$

≈ 0

penalize the coefficient
some bigger square
of slope term



overfitting reducing

too many features by increasing

Feature	Lasso Regression (L1)	Ridge Regression (L2)
Penalty	Absolute values of coefficients	Squared magnitude of coefficients
Feature Selection	Yes (sets coefficients to 0)	No (shrinks close to 0)
Solution	Sparse	Non-sparse
Robustness	Better for feature selection	Better for multicollinearity

the training time
of the
training set

Simple set
of features

Independent features/variables
are correlated
(multicollinearity)

Elastic Net

combination of both L1 &

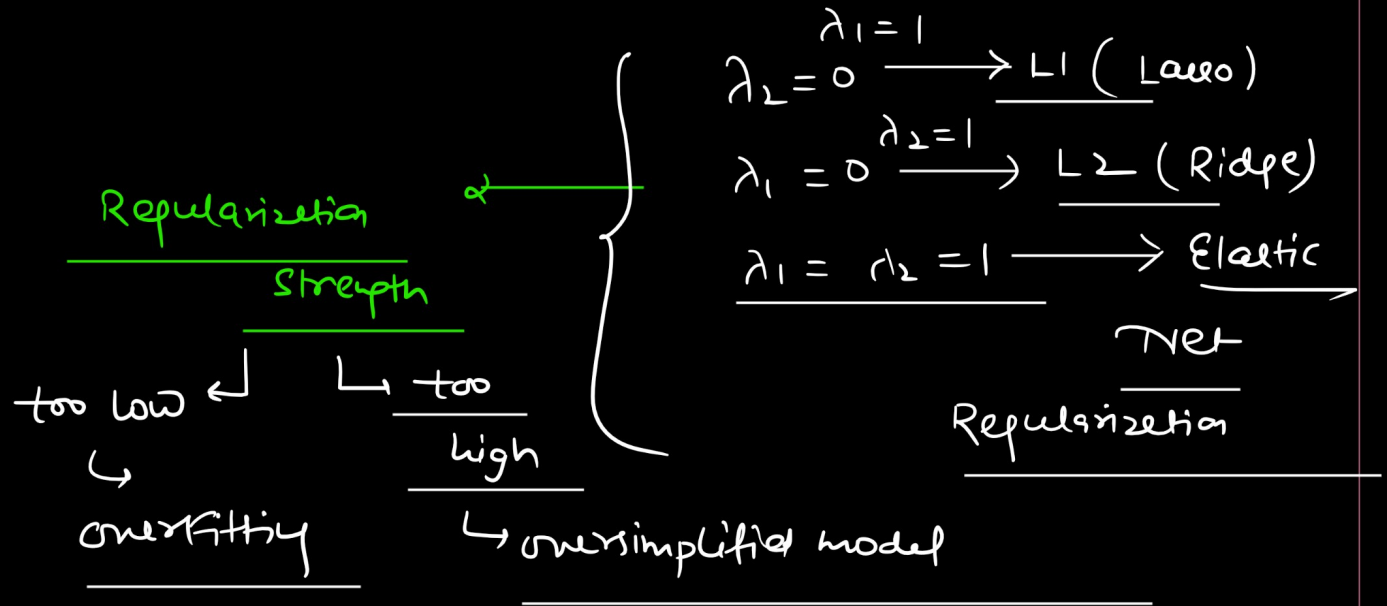
alpha in

L2 Regularization

documentation

technique

$$\text{cost function} = MSE + \lambda_1 \sum_{i=1}^n |m_i| + \lambda_2 \sum_{i=1}^n m_i^2$$



balanced value of α (λ, α)